

Enhancing Degraded Grayscale Images

CE 490 - Image Processing Project

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Outline

- 1 Problem Discovery & Analysis
- 2 Measurement Metrics
- 3 Pipeline Overview
- 4 Pipeline Steps - Detailed
 - Step 0: Noise Analysis
 - Step 1: Adaptive Median Filter
 - Step 2: Non-Local Means
 - Step 3: Frequency Domain Sharpening
 - Step 4: Laplacian Enhancement
 - Step 5: Gaussian Smoothing
 - Step 6: Final NLM
- 5 Conclusion
- 6 Image Comparisons

Histograms of Degraded and Original Images

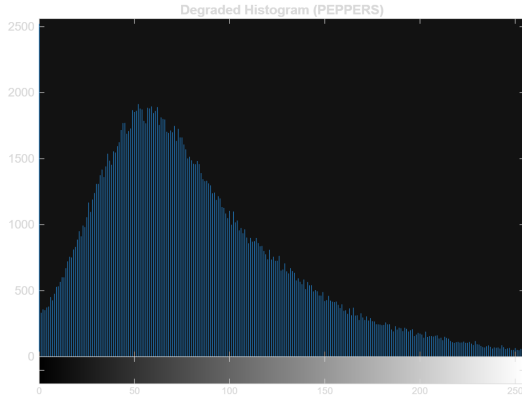


Figure: Histogram of the degraded image

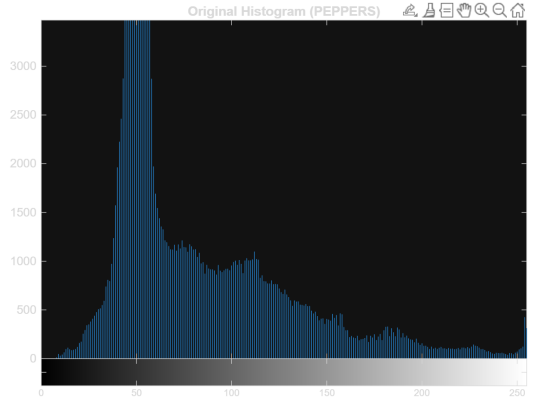


Figure: Histogram of the original image

FFTs of Degraded and Original Images

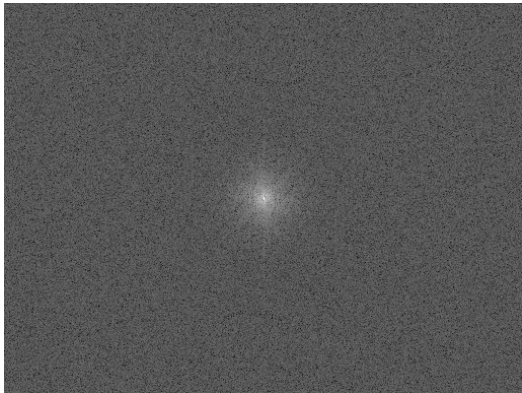


Figure: FFT spectrum of the degraded image

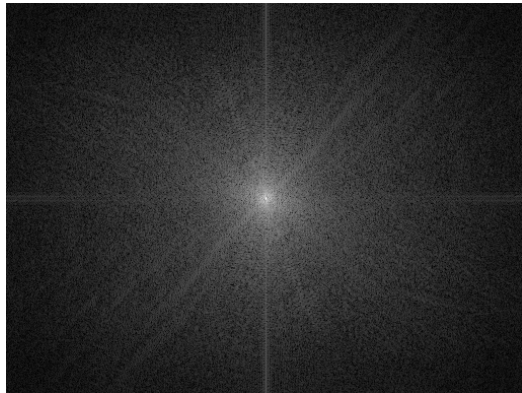
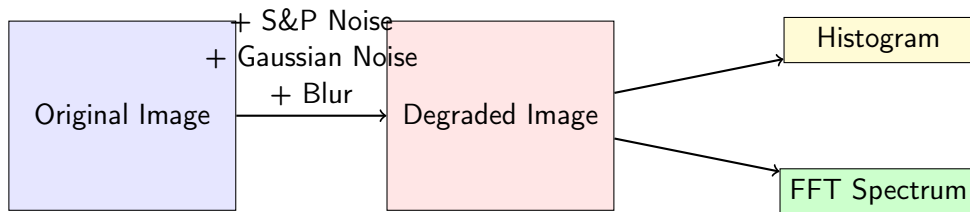


Figure: FFT spectrum of the original image

Degraded Image Analysis



Histogram Analysis:

- Spikes at 0 and 255 → S&P noise
- Broadened distribution → Gaussian noise

FFT Analysis:

- Dominant low frequencies → blur

Problem

Challenge

Images degraded by **mixed noise types**:

- **Salt & Pepper Noise** - Impulse noise causing extreme pixel values (0 or 255)
- **Gaussian Noise** - Additive gaussian noise affecting all pixels
- **Edge Degradation** - Loss of sharpness and fine details

Goal

Restore degraded images while:

- Removing salt & pepper noise
- Reducing Gaussian noise
- Preserving and enhancing edges
- Maximizing PSNR, SSIM and minimizing MSE metrics

Salt & Pepper Ratio Measurement

Method: Pixel Counting

Count pixels with extreme values (0 or 255):

$$\text{S\&P Ratio} = \frac{N_{\text{salt}} + N_{\text{pepper}}}{N_{\text{total}}}$$

Algorithm:

- 1 Count pixels = 255 (salt)
- 2 Count pixels = 0 (pepper)
- 3 Compute ratio

Usage in Pipeline:

- Initial noise estimation
- Adaptive Median termination
- Threshold: 1% (0.01)

Gaussian Noise Estimation - Laplacian-MAD Method

Principle

The Laplacian operator enhances noise while suppressing smooth regions. Using Median Absolute Deviation (MAD) provides robust estimation.

Laplacian Kernel:

$$\nabla^2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

MAD:

$$\text{MAD} = \text{median}(|\nabla^2 I|)$$

Normalization Factor: $k = 0.6745 \times \sqrt{20} \approx 3.015$
(0.6745: Gaussian MAD-to- σ , $\sqrt{20}$: Laplacian L2 norm)

Final Noise Estimation:

$$\sigma_{noise} = \frac{\text{median}(|\nabla^2 I|)}{0.6745 \times \sqrt{20}}$$

Edge Strength (Tenengrad Metric):

$$E = \frac{1}{MN} \sum \sqrt{G_x^2 + G_y^2}$$

where G_x, G_y are Sobel gradients

Flat-Region Noise:

$$N_{flat} = \text{std}(\nabla^2 I|_{flat})$$

Flat mask: bottom 30% of gradient magnitudes

MSE

$$\text{MSE} = \frac{1}{MN} \sum_{i,j} (I_o - I_r)^2$$

- Lower is better (between [0-1])

PSNR

$$\text{PSNR} = 10 \log_{10} \frac{1}{\text{MSE}}$$

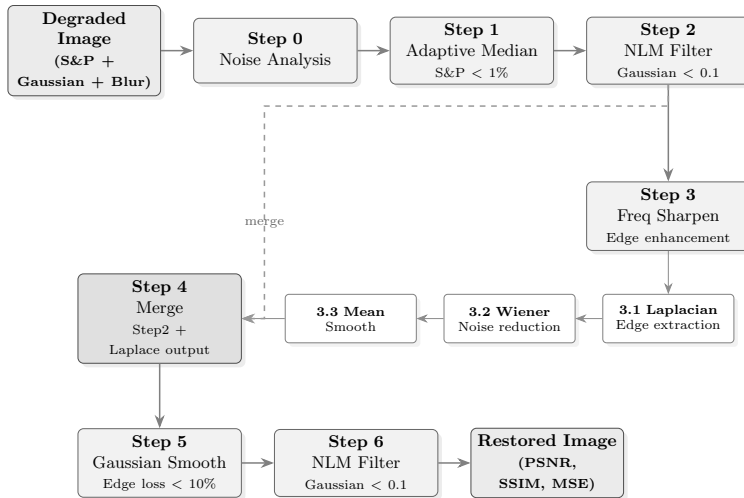
- Higher is better

SSIM

$$\text{SSIM} = f(I, c, s)$$

- Range: [-1, 1]
- 1 = identical

Complete Pipeline Architecture



Degraded Image Visualization



Step 0: Initial Noise Analysis

Degraded Image Metrics (Before Processing)

Image	MSE	PSNR (dB)	SSIM
Boat	0.0173	17.6277	0.1732
Baboon	0.0223	16.5192	0.1571
Barbara	0.0197	17.0460	0.1568
Peppers	0.0164	17.8637	0.1380
Cameraman	0.0195	17.089	0.1628

Step 1: Iterative Adaptive Median Filter

Goal

Remove S&P noise iteratively until ratio drops below 1% threshold.

Two-Level Decision with Dynamic Window Growth ($3 \times 3 \rightarrow 5 \times 5$):

Level A: Is median valid?

Level B: Is pixel valid?

$$z_{min} < z_{med} < z_{max}$$

$$z_{min} < z_{xy} < z_{max}$$

- If YES $\rightarrow$ Go to Level B
- If NO $\rightarrow$ Expand window
- If YES $\rightarrow$ Keep z_{xy} (preserve structure)
- If NO $\rightarrow$ Replace with z_{med}

Advantages

- Smaller windows in low-noise areas $\rightarrow$ better structure preservation
- Larger windows handle dense noise regions effectively

Step 1: Result After Adaptive Median



Step 1: Metrics

Metrics After Step 1 (All Images)

Image	MSE	PSNR (dB)	SSIM
Boat	0.0036	24.47	0.572
Baboon	0.0086	20.68	0.382
Barbara	0.0051	22.89	0.550
Peppers	0.0014	28.68	0.715
Cameraman	0.0061	22.12	0.570

Observation

Adaptive Median removes salt & pepper noise and affects the metrics positively.

Step 2: Iterative Non-Local Means (NLM)

Goal

Remove Gaussian noise while preserving texture and fine details.

NLM:

$$\hat{I}(p) = \sum_{q \in S} w(p, q) \cdot I(q)$$

$$w(p, q) = \frac{1}{Z(p)} \exp \left(-\frac{\|N(p) - N(q)\|^2}{h^2} \right)$$

- $N(p)$: Neighborhoods around pixel p
- S : Search window
- h : Degree of smoothing (filtering parameter)
- $Z(p)$: Normalization factor

Step 2: NLM - Grid Search Parameters

Parameter Space

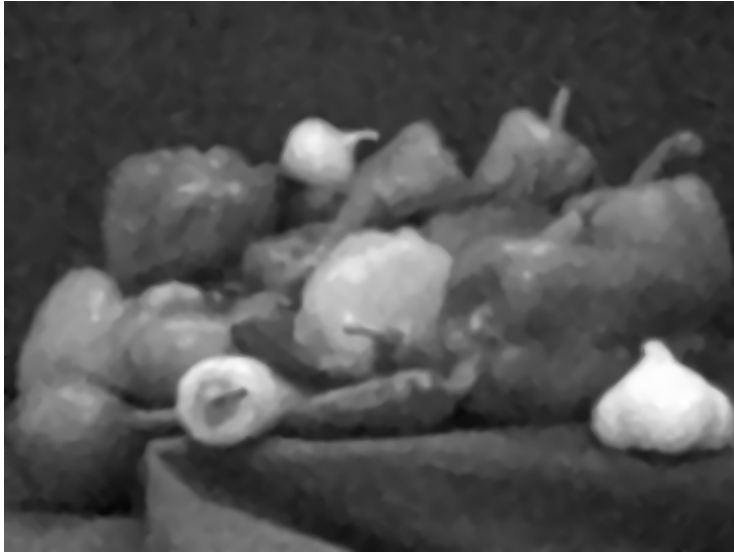
Parameter	Values
Degree of Smoothing	[5, 10, 15, 20, 30]
Search Window Size	[11, 21] (odd)
Comparison Window Size	[3, 5, 7] (odd)

Stopping Conditions:

Selection Criterion:

- Minimize estimated Gaussian noise
- Gaussian noise < 0.1
- Max iterations: 5
- No improvement found

Step 2: Result After NLM



Step 2: Results - Metrics

Metrics After Step 2 (All Images)

Image	MSE	PSNR (dB)	SSIM
Boat	0.0035	24.62	0.620
Baboon	0.0089	20.49	0.361
Barbara	0.0049	23.07	0.615
Peppers	0.0010	30.02	0.860
Cameraman	0.0059	22.27	0.679

Observation

NLM reduces Gaussian noise while preserving structural similarity.

Step 3: Iterative High-Pass Filtering

Goal

Restore edges using frequency-domain filtering.

Gaussian High-Pass:

$$H_{HP}(u, v) = 1 - H_{LP}(u, v) = 1 - e^{-\frac{D(u, v)^2}{2\sigma_c^2}}$$

where $D(u, v) = \sqrt{u^2 + v^2}$ is the distance from center.

Parameters

- k : Boost strength [0.3, 0.5, 0.7, 0.9, 1.1, 1.3, 1.5, 1.8, 2.0]
- σ_c : Cutoff frequency [0.03, 0.05, 0.07, 0.09, 0.11, 0.13]

Step 3: Sharpening - Selection Criteria

Edge vs Noise

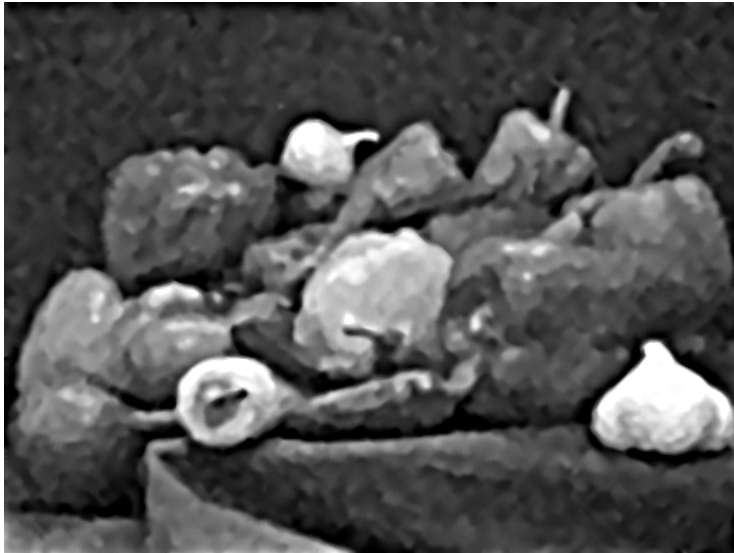
$$\text{ratio} = \frac{\text{noise_gain}}{\text{edge_gain}}$$

- $\text{edge_gain} = \frac{E_{\text{new}}}{E_{\text{old}}} - 1$ (how much edge increased)
- $\text{noise_gain} = \frac{N_{\text{new}}}{N_{\text{old}}} - 1$ (how much noise increased)
- **Select:** Parameter with **lowest ratio** (max edge, min noise)

Early Stopping Conditions

- 1 Ratio ≥ 3.0 (noise increasing too fast)
- 2 Noise level > 0.03 (absolute threshold)
- 3 Max iterations: 5

Step 3: Result After Sharpening



Step 4: Laplacian Enhancement

Goal

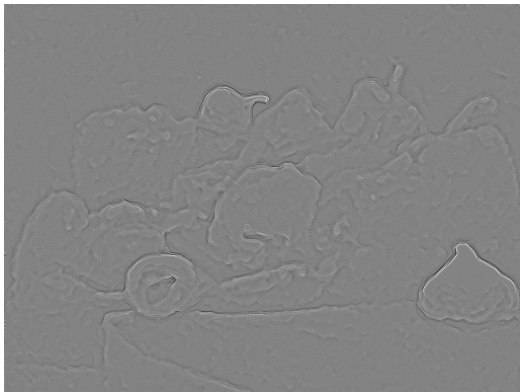
Enhance edges using Laplacian filtering on sharpened image (Step 3) and combine with NLM output.

Process:

- 1 Compute Laplacian of sharpened image (Step 3)
- 2 Apply Wiener filter to reduce noise (using estimated variance)
- 3 Apply mean filter (3×3) for smoothing the edges
- 4 Add processed Laplacian to Step 2 (NLM) output

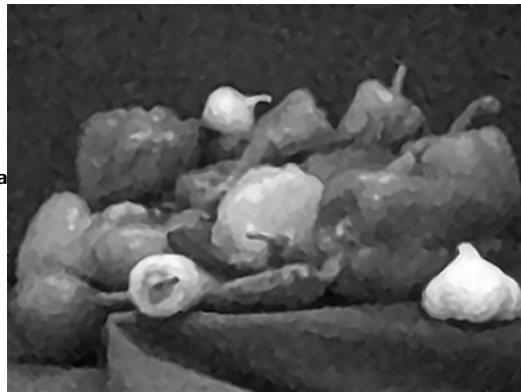
$$I_{final} = I_{NLM} + \text{mean}(\text{wiener}(\nabla^2 I_{sharpened}))$$

Step 4: Results - Images



Laplacian Edge Output

→
**Laplacian
Merge**
+ NLM
Step 2
Output



Merged Result

Step 4: Result After Merging Laplacian and NLM Output

Metrics After Step 4 (All Images)

Image	MSE	PSNR (dB)	SSIM
Boat	0.0032	24.94	0.624
Baboon	0.0084	20.76	0.396
Barbara	0.0047	23.31	0.621
Peppers	0.0010	30.11	0.826
Cameraman	0.0048	23.18	0.689

Observation

Laplacian enhancement adds edge details from Step 3 while maintaining denoised quality from Step 2.

Step 5: Iterative Gaussian Smoothing

Goal

Remove residual while preserving edges.

Gaussian Filter:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

Parameters Grid

- σ : [0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.8, 1.0]
- Filter Size: [3, 5, 7, 9, 11, 13]

Step 5: Gaussian - Selection Criterion

Edge Preservation Priority

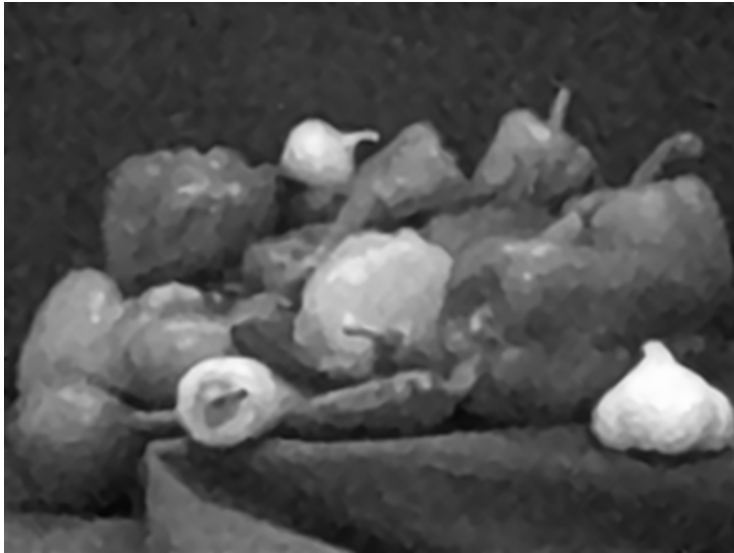
$$\text{ratio} = \frac{\text{noise_reduction}}{\text{edge_loss}}$$

Select parameter with **highest ratio** (max noise reduction per edge loss).

Iteration Conditions

- **Total edge loss** must remain $< 10\%$ from initial
- Only apply if noise actually decreases
- Early stopping when no valid parameters found
- Max iterations: 5

Step 5: Result After Gaussian Smoothing



Step 5: Results - Metrics

Metrics After Step 5 (All Images)

Image	MSE	PSNR (dB)	SSIM
Boat	0.0030	25.22	0.635
Baboon	0.0085	20.73	0.390
Barbara	0.0047	23.30	0.625
Peppers	0.0009	30.45	0.846
Cameraman	0.0049	23.12	0.693

Observation

Gaussian smoothing provides final smoothing to reduce noise while maintaining edge loss below.

Step 6: Iterative NLM Again

Goal

Apply NLM one more time to remove any noise introduced by previous enhancement steps.

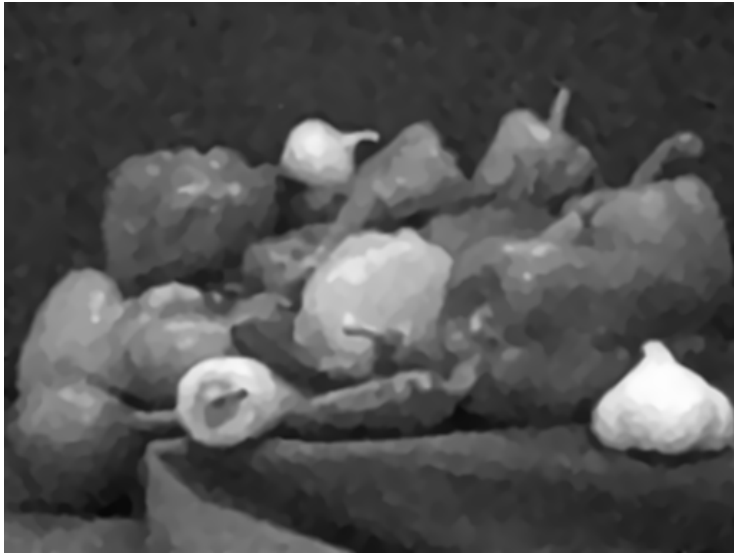
Why?

- Sharpening and Laplacian enhancement can amplify residual noise
- Use same grid search approach as Step 2

Parameters (Same as Step 2)

Parameter	Values
Degree of Smoothing	[5, 10, 15, 20, 30]
Search Window Size	[11, 21]
Comparison Window Size	[3, 5, 7]
Target Threshold	0.08 (stricter)

Step 6: Result After NLM



Step 6: Results - Metrics

Metrics After Step 6 (All Images)

Image	MSE	PSNR (dB)	SSIM
Boat	0.0029	25.41	0.648
Baboon	0.0088	20.57	0.367
Barbara	0.0047	23.31	0.629
Peppers	0.0009	30.67	0.860
Cameraman	0.0048	23.16	0.707
Global Avg	0.0044	24.63	0.642

Observation

Final NLM application removes noise amplified by enhancement steps.

What We Achieved?

- ① **Multi staged pipeline** for mixed noise removal
- ② **Adaptive parameter selection for different images** via grid search
- ③ **Iterative processing** with convergence criteria by looking at quantitative metrics

Boat - Original vs Degraded vs Restored



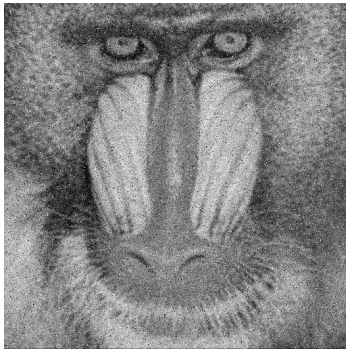
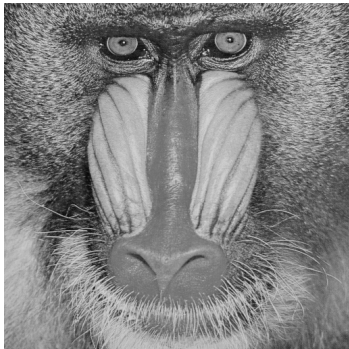
Boat - Metrics

MSE: 0.0029

PSNR: 25.41 dB

SSIM: 0.648

Baboon - Original vs Degraded vs Restored



Baboon - Metrics

MSE: 0.0088

PSNR: 20.57 dB

SSIM: 0.367

Barbara - Original vs Degraded vs Restored



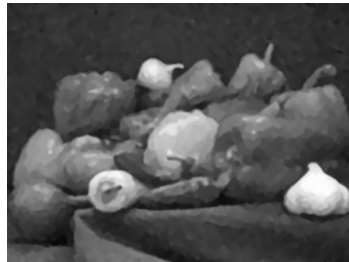
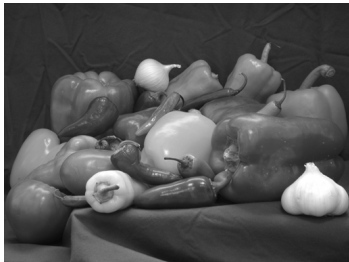
Barbara - Metrics

MSE: 0.0047

PSNR: 23.31 dB

SSIM: 0.629

Peppers - Original vs Degraded vs Restored



Peppers - Metrics

MSE: 0.0009

PSNR: 30.67 dB

SSIM: 0.860

Cameraman - Original vs Degraded vs Restored



Cameraman - Metrics

MSE: 0.0048

PSNR: 23.16 dB

SSIM: 0.707

Global Results Summary - All Images

Final Metrics

Image	MSE	PSNR (dB)	SSIM
Boat	0.0029	25.41	0.6482
Baboon	0.0088	20.57	0.3668
Barbara	0.0047	23.31	0.6289
Peppers	0.0009	30.67	0.8596
Cameraman	0.0048	23.16	0.7074
Global Average	0.0044	24.63	0.6422

Thank You For Listening!

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